Design and Simulation of Silicon Photonic Interferometers

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Seven TE circuits are laid out: six interferometers, simulated to determine free spectral range and waveguide group index, and one GC-to-GC reference.

I Theory

$$n_g = n_p - \lambda \frac{dn_p}{d\lambda}, \quad f_i = \frac{c}{\lambda_i} \quad (1)$$

$$\Delta f = \frac{1}{N-1} \Sigma |f_{i+1} - f_i|, \quad \Delta\lambda \approx \frac{\lambda^2 \Delta f}{c} \quad (2)$$

$$n_{g,MZI} = \frac{c}{\Delta L\, \Delta f}, \quad n_{g,MM} = \frac{c}{2\Delta L\, \Delta f} \quad (3)$$

ΔL is the arm-length difference; L is the straight FP cavity length. c = 299,792,458 m/s. N is the number of selected resonances. FP and MFP use the complex grating response in Section III.

II Design and simulation

Device	deviceID	Length	Bragg mirrors
MZI 30 µm	MZI1	ΔL = 29.998 µm	—
MZI 150 µm	MZI2	ΔL = 150.002 µm	—
MM 100 µm	MM	ΔL = 100 µm	300 / 300 periods, 200 nm corrugation
FP 150 µm	FP1	L = 149.999 µm	100 / 100 periods, 50 nm corrugation
FP 250 µm	FP2	L = 249.999 µm	100 / 100 periods, 50 nm corrugation
MFP 100 µm	MFP1	L = 100 µm	10 / 300 / 300 periods, 200 nm corrugation
GC-to-GC	GC_to_GC	150 µm route	—

Measurement labels follow opt_in_TE_1550_device_weizhu_<deviceID>. MM: Michelson interferometer. Silicon strip: 500 nm width, 220 nm thickness; TE polarization; 5 µm bend radius. Bragg period: 317 nm. Directional coupler: 200 nm gap, 10 µm coupling length. In the FP circuits, a 5 µm straight waveguide connects each grating coupler to its Bragg mirror.

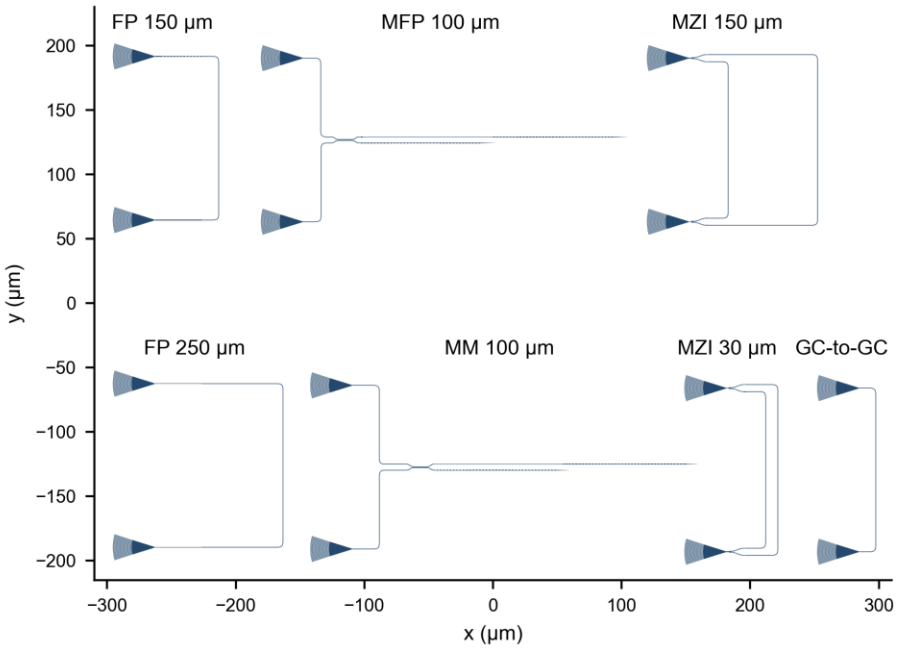


Figure 1 EBeam_weizhu.gds layout. Seven input–output pairs with 127 µm grating-coupler spacing. Optical connectivity: 45 matched pin pairs.

PDK strip model at 1550 nm: n_p = 2.44553, n_g = 4.19088. The separate MODE export wg.mat gives n_p = 2.4462 and n_g = 4.2036 at 1550 nm.

Grating-coupler reference

The GC-to-GC circuit connects two grating couplers, 127 µm apart, with a 150 µm strip waveguide and two 90° bends of 5 µm radius. Its measured spectrum gives the grating-coupler insertion loss versus wavelength, which is subtracted from the measured spectra of the other six circuits. The bend and propagation loss of this short route is small compared with the grating-coupler loss.

Simulation results

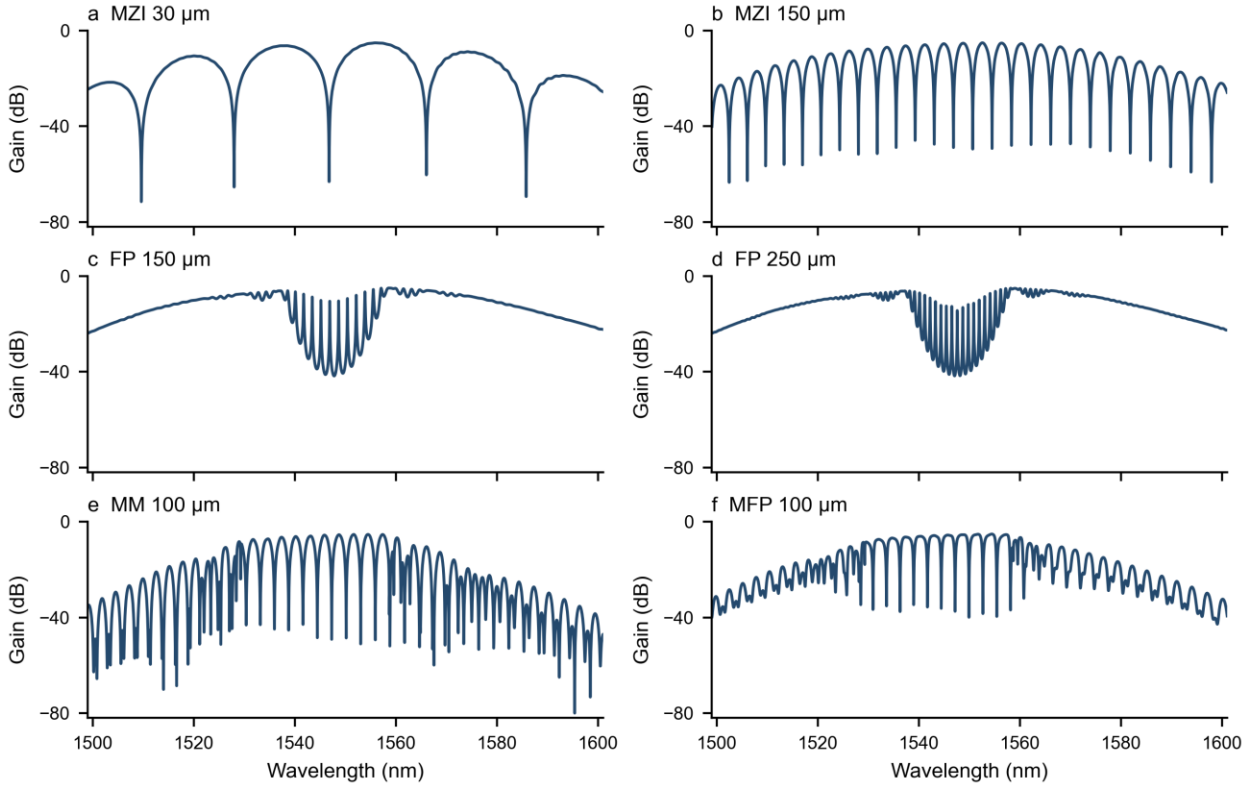


Figure 2 Saved INTERCONNECT gain spectra. 5,000 samples per device over 1499.02–1601.00 nm.

Device	FSR nm	FSR GHz	Group index
MZI 30 µm	19.03	2384.3	4.192
MZI 150 µm	3.81	476.6	4.193
MM 100 µm	2.86	357.5	4.193
FP 150 µm	1.74	217.3	4.196
FP 250 µm	1.08	135.1	4.192
MFP 100 µm	2.75	343.8	4.188

FSR uses consecutive FP transmission maxima and MZI, MM and MFP minima. Selected features: 1542–1555 nm; MZI 30: 1520–1575 nm. FP and MFP indices are fitted with the nominal PDK grating model.

Group index calculation

Device	Calculation	n_g
MZI 30	$299792458 / (29.998 \times 10^{-6} \times 2384.2721 \times 10^{9})$	4.192
MZI 150	$299792458 / (150.002 \times 10^{-6} \times 476.6223 \times 10^{9})$	4.193

Device	Calculation	n_g
MM 100	299792458 / (2 × 100 × 10^{-6} × 357.4784 × 10^{9})	4.193

$$n_{g,FP\ pair} \approx \frac{c}{2 \times 100 \times 10^{-6}} \left[\frac{1}{135.0940 \times 10^{9}} - \frac{1}{217.3032 \times 10^{9}}\right] = 4.198 \quad (4)$$

FP pair estimate assumes equal mirror delay over the selected frequency windows. FSR measures group delay; the absolute phase index n_p is supplied by the waveguide model.

Data files: TE_MZI_30um.txt, TE_MZI_150um.txt, TE_MM_100um.txt, TE_FP_150.txt, TE_FP_250.txt and TE_MFP_100um.txt.

III FP and MFP calculation

u = exp(iβL). β(f) = β₀ + 2πn_g(f − f₀)/c. r and t are complex grating reflection and transmission coefficients.

$$H_{FP} = \frac{t^2 u}{1 - r^2 u^2} \quad (5)$$

$$r_{cav} = r_f + \frac{t_f^2 r_b u^2}{1 - r'_f r_b u^2} \quad (6)$$

MFP: f = front grating; b = rear grating; r′_f = reflection from the cavity side. The front grating is 3.17 µm long; the straight cavity is 100 µm.

$$P_{FP} \propto |H_{FP}|^2, \quad P_{MFP} \propto |r_{cav} + r_{ref}|^2 \quad (7)$$

Grating model	FP mirrors	MFP front rear reference
Period	317 nm	317 nm
Corrugation	50 nm	200 nm
Periods	100 / 100	10 / 300 / 300
Bragg wavelength	1547.7045 nm	1543.8124 nm
Coupling κ	+75375.25 m^{-1}	−159056 m^{-1}
Loss	590 dB/m	590 dB/m

Complex r and t use the uniform coupled-mode grating response with these nominal EBeam model parameters. n_g is fitted freely to all 638 samples at 1542–1555 nm, with a constant phase offset, smooth gain envelope and small power floor.

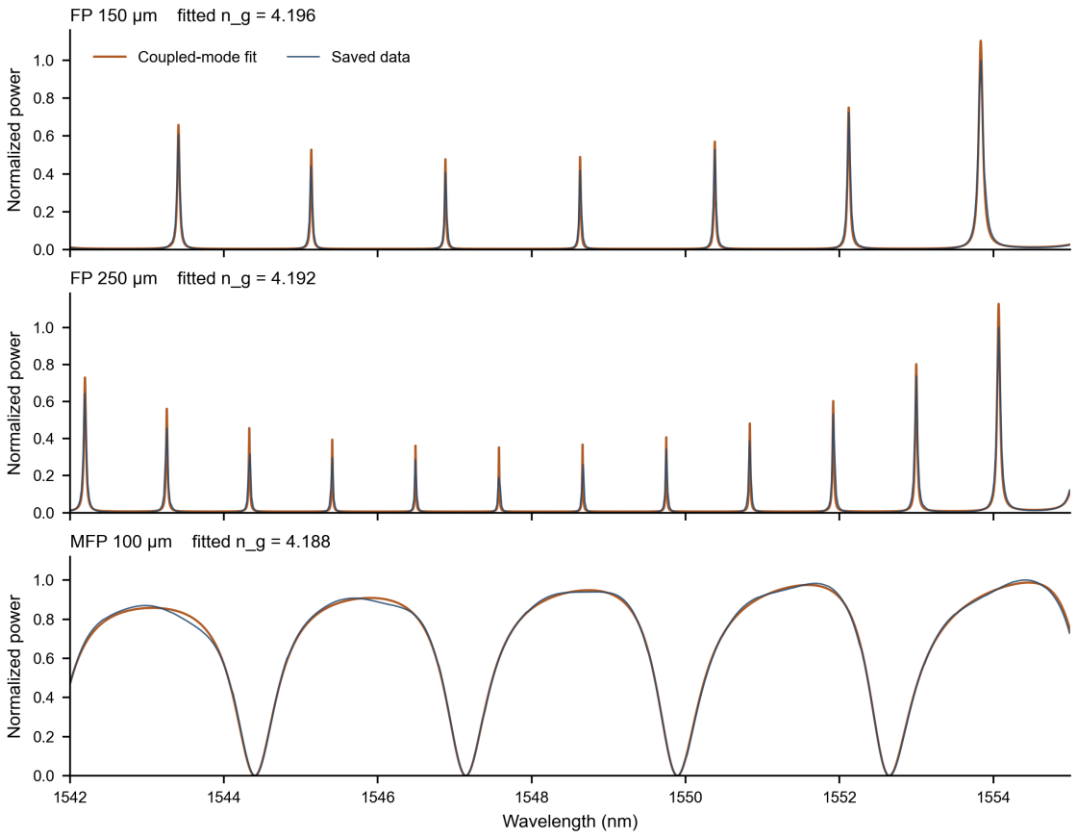


Figure 3 Saved power and fitted response. RMS residual relative to the maximum saved power in this window: FP 150, 1.29%; FP 250, 1.93%; MFP 100, 1.05%.

Model sources: installed EBeam ebeam_bragg_te1550.ice and WaveGuideTETMStrip,w=500,h=220.txt; Ansys INTERCONNECT Bragg Grating WBG element documentation.